JUnit Exercise

Module 4 - JUnit Basic Testing Exercises

Exercise 1 - Calculator.java

```
package com.cognizant.junit;

public class Calculator {
    public int add(int a,int b){ return a+b; }
    public int subtract(int a,int b){ return a-b; }
    public int multiply(int a,int b){ return a*b; }
    public int divide(int a,int b){ return a/b; }
}
```

Exercise 2 - CalculatorTest.java

```
package com.cognizant.junit;

import static org.junit.Assert.*;
import org.junit.Test;

public class CalculatorTest {
    Calculator calculator = new Calculator();

    @Test public void testAddition(){ assertEquals(15, calculator.add(10,5)); }
    @Test public void testSubtraction(){ assertEquals(5, calculator.subtract(10,5)); }
    @Test public void testMultiplication(){ assertEquals(50, calculator.multiply(10,5)); }
    @Test public void testDivision(){ assertEquals(2, calculator.divide(10,5)); }
}
```

Exercise 3 - AssertionsTest.java

```
package com.cognizant.junit;

import static org.junit.Assert.*;
import org.junit.Test;

public class AssertionsTest {
    @Test
    public void testAssertions(){
        assertEquals(5,2+3);
        assertTrue(10>5);
        assertFalse(5>10);
        assertNull(null);
        assertNotNull(new Object());
        String str="JUnit";
        assertSame(str,str);
        assertNotSame(new String("Java"), new String("Java"));
```

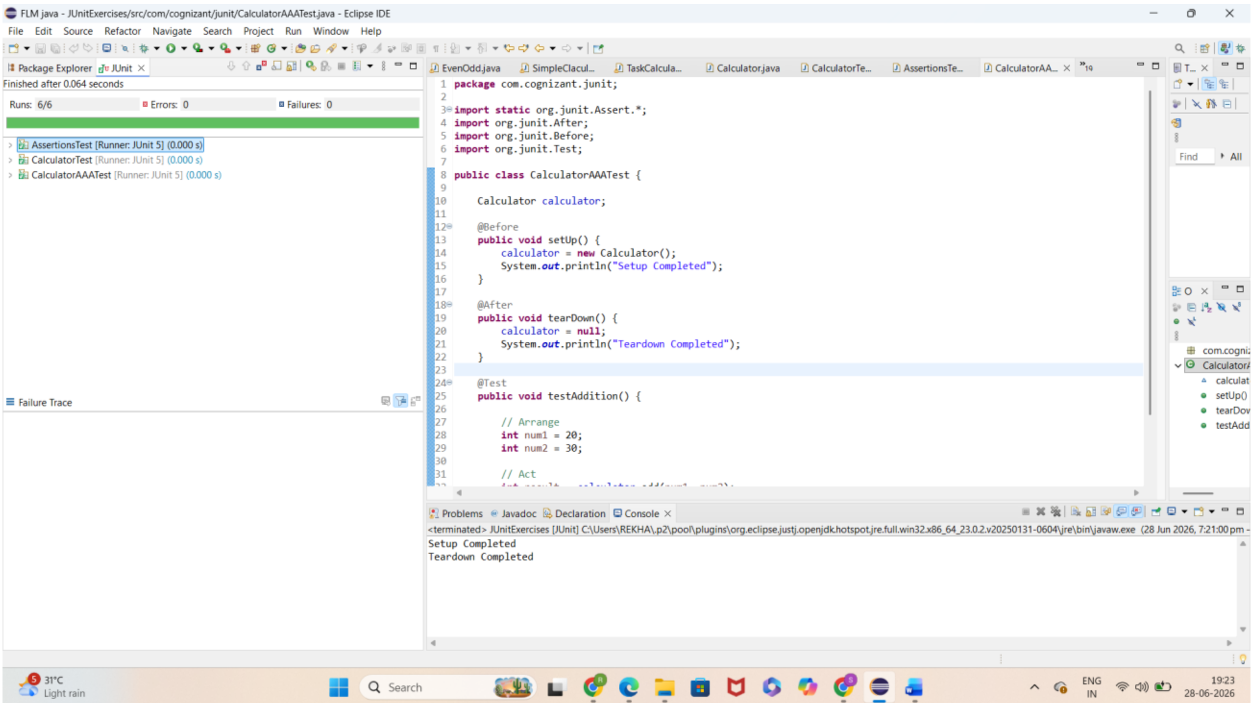
```
    }
}
```

Exercise 4 - CalculatorAAATest.java

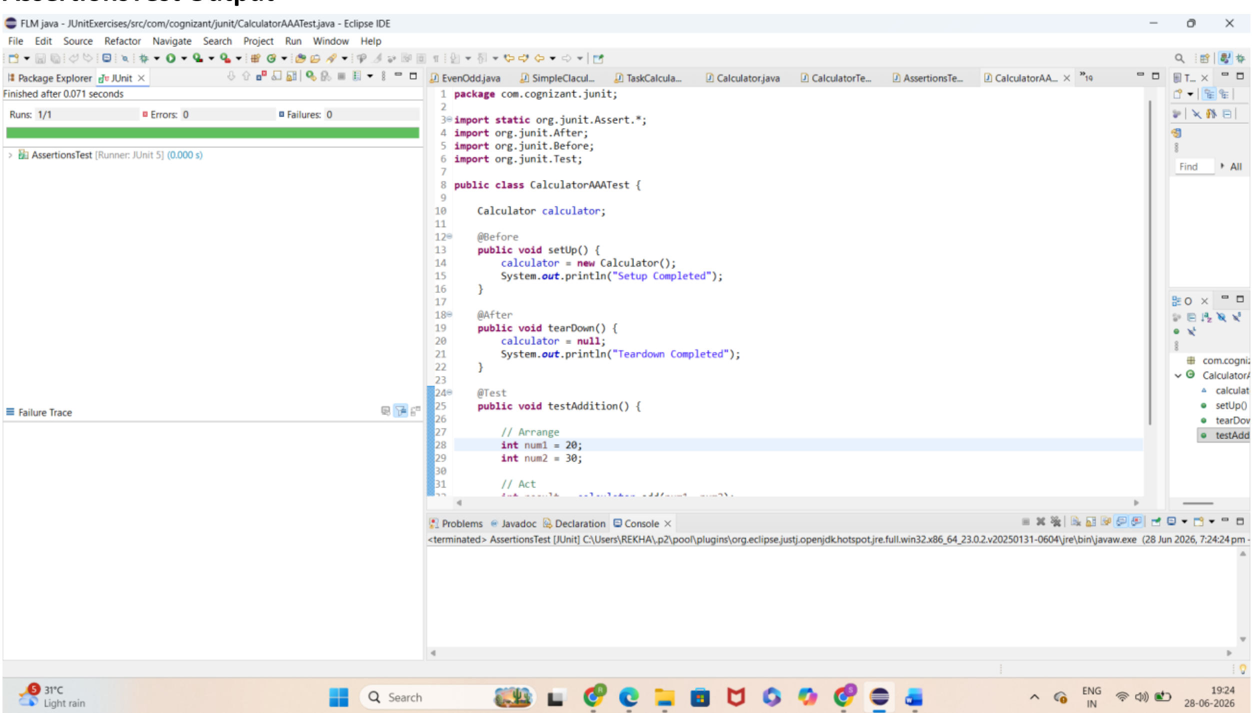
```
package com.cognizant.junit;

import static org.junit.Assert.*;
import org.junit.Before;
import org.junit.After;
import org.junit.Test;

public class CalculatorAAATest {
    Calculator calculator;
    @Before
    public void setUp(){
        calculator=new Calculator();
        System.out.println("Setup Completed");
    }
    @After
    public void tearDown(){
        calculator=null;
        System.out.println("Teardown Completed");
    }
    @Test
    public void testAddition(){
        int num1=20, num2=30;
        int result=calculator.add(num1,num2);
        assertEquals(50,result);
    }
}
```

Execution Output Screenshots

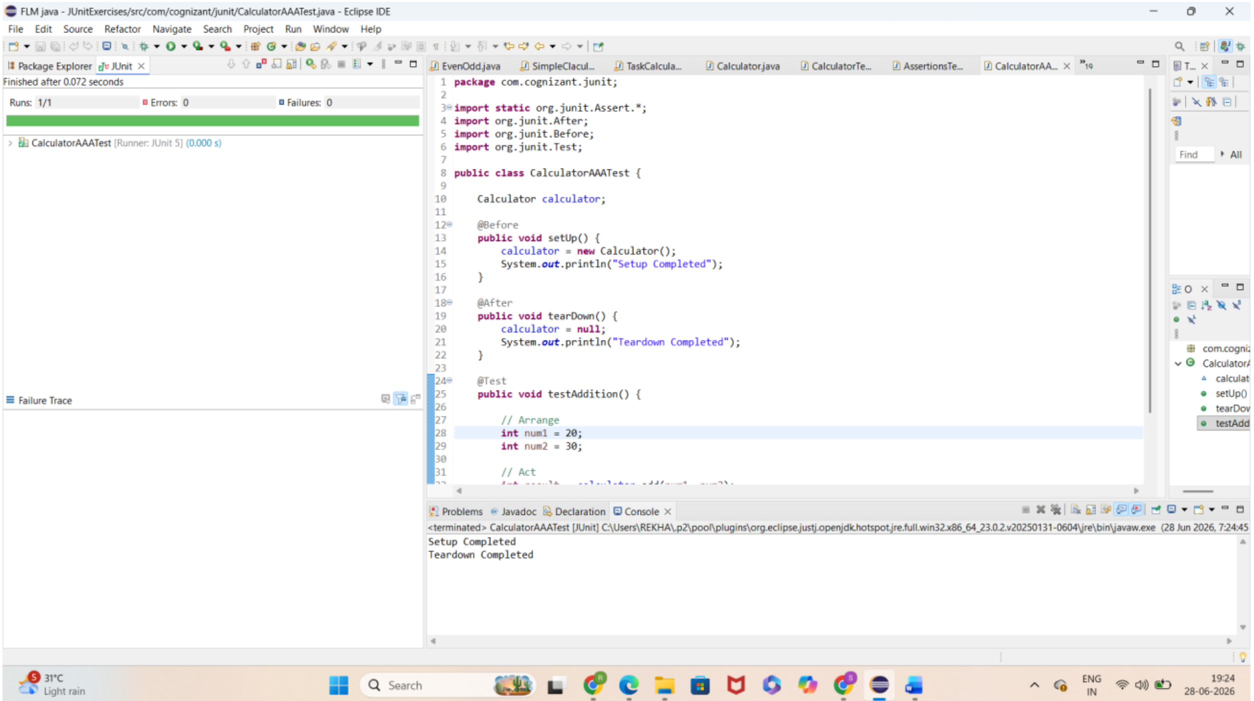
Run All Tests



AssertionsTest Output



CalculatorAAATest Output



CalculatorTest Output